

Reproduce-then-Extend — State Welfare Spending (American State Politics) (Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. The determinants of **state welfare effort** are a long-running question in American state politics and policy (e.g., Plotnick & Winters, “A Politico-Economic Theory of Income Redistribution,” *APSR* 1985; Brown, “State Responses to the Welfare Crisis,” 1995). The canonical design is a **cross-sectional OLS** of per-capita state welfare spending on economic need, fiscal capacity, political ideology, and organized labor. This example illustrates **regularization and variable selection** in a small-sample ($n = 50$) design.

1 Background

Why do some states spend far more on social welfare than others? The state welfare-effort literature points to **fiscal capacity** (richer states can afford more), **need** (poverty and inequality), **political ideology and party control** (liberal, Democratic states redistribute more), and **organized labor**. These explanations are tested with ordinary cross-state regressions. Here we take the **cross-section of the 50 states** and ask which of these characteristics predict welfare spending – and whether penalized regression predicts new states better than ordinary least squares.

2 Data and codebook

Unit of analysis: a U.S. **state** (cross-section averaged over 2005–2012); $n = 50$. Data come from the **Correlates of State Policy Project** (Jordan & Grossmann 2020).

Variable	Definition
welfare_spending_percap	per-capita state public-welfare spending (USD) (outcome)
mood	public policy mood (higher = more liberal preferences)
citi6013	citizen ideology (Berry et al.), higher = more liberal
ranney4_control	Ranney index of Democratic party control (higher = more Democratic)
union_density	percent of workers unionized
evangelical_pop	evangelical share of the population
gini_coef	income inequality (Gini coefficient)
gsppcap	gross state product per capita (fiscal capacity)

3 Descriptive results

```

d <- read.csv("data/state_welfare_spending.csv")
preds <- setdiff(names(d), c("st", "welfare_spending_percap"))
cat(sprintf("%d states x %d predictors (p/n = %.2f)\n", nrow(d), length(preds),
  ↪ length(preds)/nrow(d)))

## 50 states x 7 predictors (p/n = 0.14)

round(t(sapply(d[c("welfare_spending_percap", preds)],
  function(x) c(mean = mean(x), sd = sd(x), min = min(x), max = max(x)))),
  ↪ 2)

##               mean      sd      min      max
## welfare_spending_percap 1229.66 342.78  610.91 2237.38
## citi6013                53.36  14.70  23.34  86.52
## ranney4_control          0.49   0.31   0.00   1.00
## evangelical_pop          19.55  13.24   2.90  71.40
## union_density            11.23   5.60   3.14  24.91
## gini_coef                 0.61   0.03   0.56   0.68
## mood                     41.23   5.96  31.14  63.17
## gsppcap                  41596.20 7747.83 28955.83 63450.17

par(mfrow = c(1, 2)); par(mar = c(4, 4, 2.5, 1))
hist(d$welfare_spending_percap, breaks = 12, col = "#a6cee3", border = "white",
  main = "Outcome: welfare spending", xlab = "per-capita welfare spending")
cors <- sort(sapply(d[preds], function(x) cor(x, d$welfare_spending_percap)))
par(mar = c(4, 7, 2.5, 1))
barplot(cors, horiz = TRUE, las = 1, cex.names = 0.85,
  col = ifelse(cors > 0, "#1f78b4", "#e31a1c"), main = "Correlation with spending",
  xlab = "correlation"); abline(v = 0, col = "grey50")

```

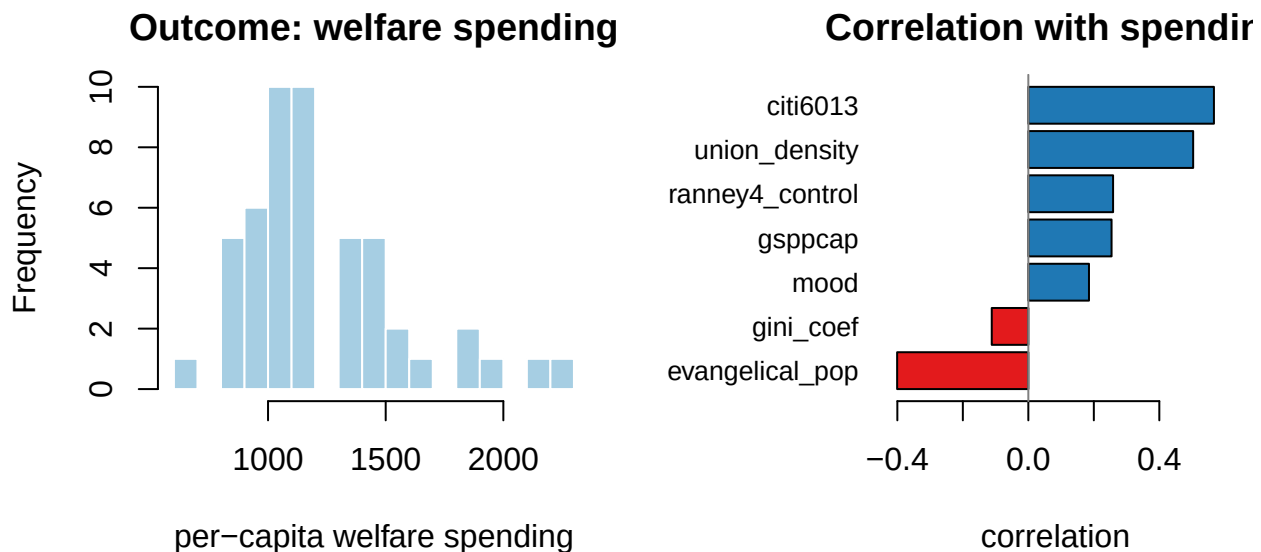


Figure 1: Welfare-spending distribution (left) and each predictor's correlation with spending (right).

4 Reproduce the published regression

The welfare-effort model regresses per-capita spending on need, capacity, ideology, and labor.

```
fit <- lm(welfare_spending_percap ~ ., data = d[, c("welfare_spending_percap", preds)])
round(summary(fit)$coefficients, 3)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    241.066    1021.345   0.236   0.815
## citi6013        15.957      5.196   3.071   0.004
## ranney4_control -85.334    158.500  -0.538   0.593
## evangelical_pop   5.261      5.093   1.033   0.307
## union_density    17.251      9.985   1.728   0.091
## gini_coef       385.886    1439.021   0.268   0.790
## mood            -15.687      8.867  -1.769   0.084
## gsppcap          0.007      0.007   1.072   0.290
```

Confirmation against the literature. In-sample $R^2 = 0.41$. Consistent with the welfare-effort literature, states with **higher income/fiscal capacity**, more liberal **ideology** and **Democratic control**, and stronger **unions** spend more per capita on welfare.

5 Regularization & variable selection

With **50 states** and **7 predictors**, ordinary least squares over-fits. Penalized regression – lasso (L_1), ridge (L_2), elastic net – shrinks the coefficients to reduce variance and improve prediction on new states.

```
X <- as.matrix(d[, preds]); y <- d$welfare_spending_percap
set.seed(2026)
lasso <- cv.glmnet(X, y, alpha = 1); ridge <- cv.glmnet(X, y, alpha = 0); enet <-
  ↪ cv.glmnet(X, y, alpha = 0.5)
n_keep <- sum(coef(lasso, s = "lambda.min")[-1] != 0)
cat(sprintf("At lambda.min the lasso keeps %d of %d predictors (the rest are set to
  ↪ exactly 0);\n",
            n_keep, length(preds)))
```

```
## At lambda.min the lasso keeps 2 of 7 predictors (the rest are set to exactly 0);
```

```
cat("the signal here is concentrated, so the sparser lambda.1se rule shrinks all the way
  ↪ to the mean.\n")
```

```
## the signal here is concentrated, so the sparser lambda.1se rule shrinks all the way to
  ↪ the mean.
```

```
data.frame(
  OLS      = round(coef(fit)[-1], 2),
  Lasso     = round(as.numeric(coef(lasso, s = "lambda.min"))[-1], 2),
  Ridge     = round(as.numeric(coef(ridge, s = "lambda.min"))[-1], 2),
  ElasticNet = round(as.numeric(coef(enet, s = "lambda.min"))[-1], 2),
  row.names = preds)
```

```
##              OLS Lasso Ridge ElasticNet
## citi6013      15.96  8.00  4.82         7.31
## ranney4_control -85.33  0.00 28.53         0.00
## evangelical_pop   5.26  0.00 -1.75         0.00
## union_density    17.25 10.33 10.09        10.94
## gini_coef       385.89  0.00 12.87         0.00
## mood           -15.69  0.00 -1.79         0.00
## gsppcap         0.01  0.00  0.00         0.00
```

Which fits best out of sample? With $n = 50$ we repeat a **70/30 split 50 times**: each replicate tunes

the penalty by cross-validation on the training states and scores once on the held-out states (penalized models at `lambda.min`).

```
rmse <- function(p, yt) sqrt(mean((yt - p)^2))
REPS <- 50; alphas <- c(Lasso = 1, Ridge = 0, ElasticNet = 0.5)
M <- matrix(NA, REPS, 4, dimnames = list(NULL, c("OLS", names(alphas))))
Xdf <- as.data.frame(X)
for (r in 1:REPS) {
  set.seed(2025 + r)
  te <- sample(nrow(X), round(0.30 * nrow(X))); tr <- setdiff(seq_len(nrow(X)), te)
  ols <- lm(y[tr] ~ ., data = Xdf[tr, ]); M[r, "OLS"] <- rmse(predict(ols, Xdf[te, ]),
  ↪ y[te])
  for (a in names(alphas)) {
    cv <- cv.glmnet(X[tr, ], y[tr], alpha = alphas[a])
    M[r, a] <- rmse(as.numeric(predict(cv, X[te, ], s = "lambda.min")), y[te])
  }
}
round(colMeans(M)) # mean held-out RMSE ($) over 50
↪ splits
```

```
##      OLS      Lasso      Ridge ElasticNet
##      350      321      334      323
```

```
best <- names(which.min(colMeans(M)[names(alphas)]))
cat(sprintf("\nBest penalization: %s. Held-out RMSE: %s = %s vs OLS %s (%.0f%%
  ↪ lower).\n",
    best, best, format(round(min(colMeans(M)[names(alphas)])), big.mark = ","),
    format(round(colMeans(M)[names(alphas) == "OLS"]), big.mark = ","),
    100 * (colMeans(M)[names(alphas) == "OLS"] - min(colMeans(M)[names(alphas)])) /
    ↪ colMeans(M)[names(alphas) == "OLS"])
```

```
##
## Best penalization: Lasso. Held-out RMSE: Lasso = 321 vs OLS 350 (8% lower).
```

With so few states per predictor, unpenalized OLS over-fits and the penalized models **predict new states' welfare spending more accurately**, while the lasso keeps a compact, interpretable set of drivers.

6 Best-subset selection with ABESS

```
library(abess)
ab <- abess(X, y)
cat(sprintf("ABESS selects a best subset of %d of %d predictors:\n", ab$best.size,
  ↪ length(preds)))
```

```
## ABESS selects a best subset of 1 of 7 predictors:
```

```
cf <- as.numeric(coef(ab, support.size = ab$best.size))[-1]
data.frame(predictor = preds, coef = round(cf, 2))[order(-abs(cf)), ]
```

```
##      predictor  coef
## 1      citi6013 13.22
## 2 ranney4_control 0.00
## 3 evangelical_pop 0.00
## 4    union_density 0.00
## 5      gini_coef 0.00
## 6         mood    0.00
```

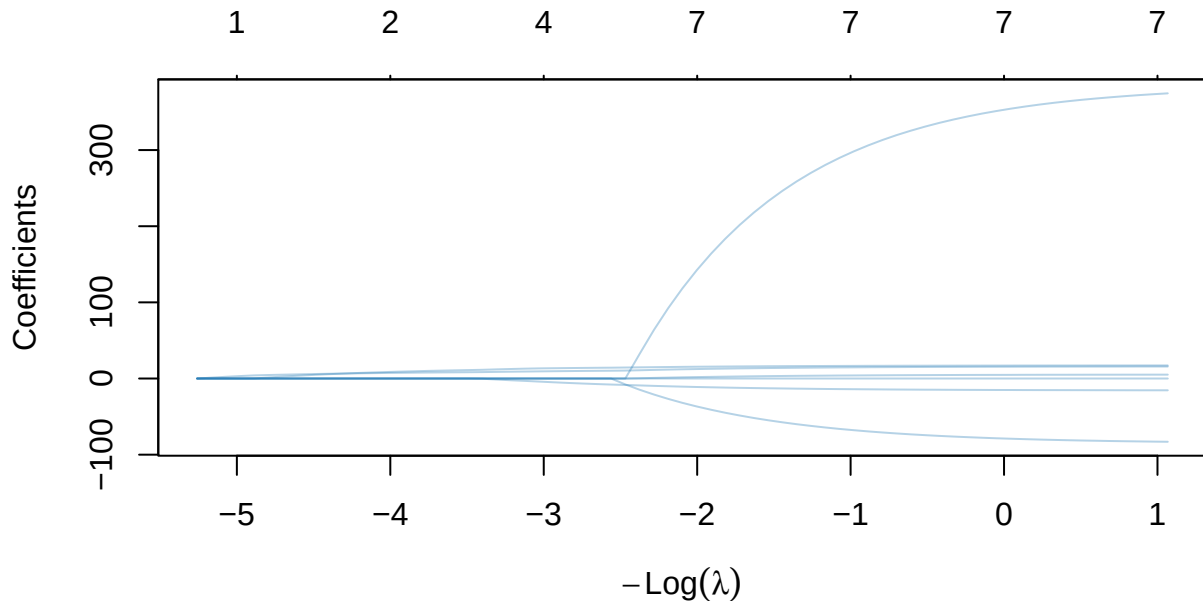


Figure 2: Lasso coefficient paths vs the penalty (log lambda). The dashed line marks the CV-optimal penalty.

```
## 7          gsppcap  0.00
```

Best subset and the lasso agree on the core drivers of welfare effort – fiscal capacity, ideology/party control, and labor – reached by direct search rather than shrinkage.

7 Takeaway

Like the state policy-liberalism example, this illustrates **regularization in a small-sample design**: with 50 states and several predictors, ordinary regression over-fits, and lasso/elastic-net (`glmnet`) plus best-subset selection (`abess`) predict new states more accurately while naming the compact set of drivers of state welfare spending.

8 Recommended exercises

1. Add more Correlates-of-State-Policy predictors and watch the held-out gain over OLS grow with the predictor-to-state ratio.
2. Log-transform the spending outcome and re-run: does the ranking of methods change?
3. Compare the lasso-selected predictors here with those selected for policy liberalism (tutorial 11) – which state characteristics matter for both?